

Employee Attrition Analysis, ABC Manufacturing Ltd

AnalystLab Africa | Data Science Internship Programme, Week 1

Project: Business Understanding & Data Exploration

Overview

This project investigates employee attrition at ABC Manufacturing Ltd on behalf of AnalystLab Africa Consulting. The goal of this phase is to understand the business problem, explore the workforce dataset, and generate evidence-based insights before any predictive modelling is considered.

Dataset

[IBM HR Analytics – Employee Attrition & Performance](#) (Kaggle), 1,470 employee records, 35 features.

Business Questions

1. What does the company's workforce look like?
2. Which departments have the highest employee attrition?
3. Does age influence attrition?
4. Does monthly income affect retention?
5. Does overtime influence attrition?
6. Which job roles experience the highest turnover?

7. Which variables appear important for future predictive modelling?

Repository Contents

File	Description
<code>attrition_analysis.ipynb</code>	Full Jupyter notebook, data loading, inspection, and visualisations
<code>Business_Understanding_Report.docx</code>	Business context, impact of attrition, and rationale for this engagement
<code>Dataset_Inspection_Report.docx</code>	Data structure, quality checks, and descriptive statistics
<code>Data_Visualisation_Report.docx</code>	Charts and interpretations (bar, histogram, boxplot, pie)
<code>Business_Insights_Report.docx</code>	Key workforce observations and attrition risk factors

File	Description
Reflection_Report.docx	Personal reflection on the week's learning
WA_Fn-UseC_-HR-Employee-Attrition.csv	Source dataset

Key Findings

- **Overtime** is one of the strongest attrition drivers: 30.5% attrition among employees who work overtime vs. 10.4% among those who don't.
- **Income** matters: the lowest income quartile has a 29.3% attrition rate vs. 10.3% in the highest quartile.
- **Age** matters: employees aged 18–25 leave at 34.8%, compared to just 9.2% for the 36–45 bracket.
- **Job role** matters: Sales Representatives have the highest attrition rate (39.8%) of any role.
- **Department**: Sales has the highest departmental attrition rate (20.6%), ahead of HR (19.0%) and R&D (13.8%).

Tools Used

Python, Pandas, Matplotlib, Seaborn, Google Colab

Author

Priscilla Arhin, Data Science Intern, AnalystLab Africa

Week 2: Feature Engineering & Data Preprocessing for Machine Learning

Building on Week 1, this phase cleans, transforms, encodes, scales, and engineers features to produce a machine-learning-ready dataset for Week 3 model development.

Additional Files

File	Description
<code>Week2_Feature_Engineering.ipynb</code>	Full preprocessing notebook (cleaning → feature engineering → encoding → scaling)
<code>Week2_Business_Understanding_Report.docx</code>	Why preprocessing matters for this engagement
<code>Data_Preprocessing_Report.docx</code>	Full documentation of the preprocessing pipeline
<code>cleaned_dataset.csv</code>	Cleaned dataset (1470)

File	Description
	× 31), constant/ID columns removed, still human- readable
<code>ml_ready_dataset.csv</code>	Machine- learning-ready dataset (1470 × 53), encoded and scaled

Pipeline Summary

Stage	Shape	Notes
Raw dataset	1470 × 35	As downloaded from Kaggle
Cleaned dataset	1470 × 31	Dropped 3 constant columns + 1 ID column
Feature-engineered	1470 × 36	Added 5 new derived features
ML-ready (final)	1470 × 53	One-hot encoding expanded categoricals; numeric features standardised

Engineered Features

- **AverageSatisfaction**, composite of 4 related satisfaction scores

- **TenureRatio**, $\text{YearsAtCompany} / \text{TotalWorkingYears}$
- **IncomePerExperienceYear**, $\text{MonthlyIncome} / \text{TotalWorkingYears}$
- **AgeGroup**, age bracket matching Week 1's EDA (18-25, 26-35, 36-45, 46-55, 56-60)
- **PromotionStagnationRatio**, $\text{YearsSinceLastPromotion} / \text{YearsAtCompany}$